

Dialogue Systems

Corpus-based Conversational Chatbots

Dr. Saeedeh Momtazi



دانشگاه صنعتی امیرکبیر
(پلی تکنیک تهران)



آکادمی‌ها



➤ Rule-base Models

- ▶ ELIZA

- ▶ PARRY

➤ Corpus-based

- ▶ Retrieval Methods

- ▶ Generation Methods



Two Architectures for Corpus-based Chatbots

➤ Response by retrieval

- ▶ Use information retrieval to grab a response (that is appropriate to the context) from some corpus

➤ Response by generation

- ▶ Use a language model or encoder-decoder to generate the response given the dialogue context



Corpus-based chatbots require corpora

- Modern corpus-based chatbots are very data-intensive
- They commonly require hundreds of millions or billions of words



What conversations to draw on?

- **Transcripts of telephone conversations between volunteers**
 - ▶ Switchboard corpus of American English telephone conversations
- **Movie dialogue**
 - ▶ Various corpora of movie subtitles
- **Hire human crowdworkers to have conversations**
 - ▶ Topical-Chat 11K crowdsourced conversations on 8 topics
 - ▶ EmpatheticDialogues 25K crowdsourced conversations grounded in a situation where a speaker was feeling a specific emotion



What conversations to draw on?

- Pre-training on larger corpora
- Pseudo-conversations from public posts on social media
 - ▶ Drawn from Twitter, Reddit, Weibo (微博), etc.
 - ▶ Tend to be noisy; often used just as pre-training.
- Crucial to remove personally identifiable information (PII)



➤ Rule-base Models

- ▶ ELIZA

- ▶ PARRY

➤ **Corpus-based**

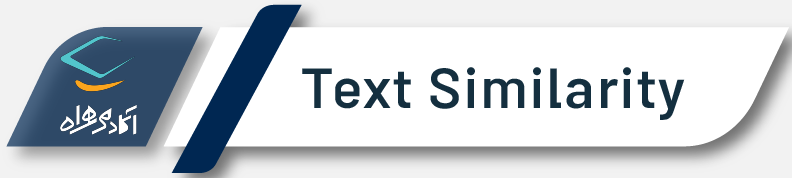
- ▶ **Retrieval Methods**

- ▶ **Generation Methods**



Response by Retrieval

- 1- Given a user turn q , and a training corpus C of conversation
 - 2- Find in C the turn r that is most similar (text similarity) to q
- Corpus-based chatbot algorithms thus draw on algorithms for question answering systems, which similarly focus on single responses while ignoring longer-term conversational goals.



➤ Classic: TF-IDF

➤ Neural Text Matching



Response by Retrieval: Classic IR Method

- 1- Given a user turn q , and a training corpus C of conversation
- 2- Find in C the turn r that is most similar (tf-idf cosine) to q
- 3- Say r

$$\text{response}(q, C) = \operatorname{argmax}_{r \in C} \frac{q \cdot r}{|q||r|}$$



Vector Space Model for Retrieval

- Providing a relevance ranking of the documents with respect to a query
 - ▶ Assign a score to each query-document pair, say in $[0, 1]$.
 - ▶ This score measures how well document and query “match”.
 - ▶ Sort documents according to scores



Vector Space Model

- Represent a doc/query by a term vector
 - ▶ Term: basic concept, e.g., word
 - ▶ Each term defines one dimension
 - ▶ N terms define an N-dimensional space
 - ▶ Query vector: $q=(x_1, \dots, x_N)$, x_i is query term weight
 - ▶ Doc vector: $d=(y_1, \dots, y_N)$, y_j is doc term weight

$$\text{relevance}(q,d) \approx \text{similarity}(q,d) = f(q,d)$$



Vector Space Model

➤ Main items in calculating scores

▶ The importance of the term in query and document:

▶ How many times does a query term occur in q and d ? → Term Frequency (TF)

▶ The general importance of the term in the collection:

▶ Is it a frequent or rare term? how often do we see the query term in the entire collection? → Document Frequency (DF)

▶ Normalization of the scores based on the length of the document:

▶ How long is d ? → Document length ($|d|$)



Term Frequency (TF)

	Anthony and Cleopatra	Julius Caesar	The Tempest	Hamlet	Othello	Macbeth	...
ANTHONY	1	1	0	0	0	1	
BRUTUS	1	1	0	1	0	0	
CAESAR	1	1	0	1	1	1	
CALPURNIA	0	1	0	0	0	0	
CLEOPATRA	1	0	0	0	0	0	
MERCY	1	0	1	1	1	1	
WORSER	1	0	1	1	1	0	
...							

	Anthony and Cleopatra	Julius Caesar	The Tempest	Hamlet	Othello	Macbeth	...
ANTHONY	157	73	0	0	0	1	
BRUTUS	4	157	0	2	0	0	
CAESAR	232	227	0	2	1	0	
CALPURNIA	0	10	0	0	0	0	
CLEOPATRA	57	0	0	0	0	0	
MERCY	2	0	3	8	5	8	
WORSER	2	0	1	1	1	5	
...							



Term Frequency (TF)

- The term frequency $tf_{t,d}$ of term t in document d is defined as the number of times that t occurs in d .
- Using raw tf for computing query-document match scores, however, is not appropriate, because
 - ▶ A document with $tf = 10$ occurrences of the term is more relevant than a document with $tf = 1$ occurrence of the term.
 - ▶ But not 10 times more relevant.
 - ▶ i.e, Relevance does not increase proportionally with term frequency.

➔ Raw Term Frequency → Log Term Frequency

$$w_{t,d} = \begin{cases} 1 + \log_{10} tf_{t,d} & \text{if } tf_{t,d} > 0 \\ 0 & \text{otherwise} \end{cases}$$



Document Frequency

➤ Rare terms are more informative than frequent terms

- ▶ Consider a term in the query that is rare in the collection (e.g., "arachnocentric").
- ▶ A document containing this term is very likely to be relevant.
- ▶ We want high weights for rare terms like "arachnocentric"

➤ Frequent terms are less informative than rare terms.

- ▶ Consider a term in the query that is frequent in the collection (e.g., "good", "increase", "line").
 - ▶ A document containing this term is more likely to be relevant than a document that doesn't, but these words are not sure indicators of relevance.
 - ▶ We want positive weights for such words, but lower weights than for rare terms.
- Using document frequency to factor this into computing the matching score.



IDF Example

➤ Compute idf_t using the formula: $\text{idf}_t = \log_{10} (1,000,000 / \text{df}_t)$

term	df_t	idf_t
calpurnia	1	6
animal	100	4
sunday	1000	3
fly	10,000	2
under	100,000	1
the	1,000,000	0



TF-IDF Weighting

- tf-idf weighting is one of the best known weighting scheme in information retrieval
- tf-idf weight of a term is the product of its tf weight and its idf weight.

$$w_{t,d} = (1 + \log tf_{t,d}) \cdot \log(N / df_t)$$

- Increases with the number of occurrences within a document. (term frequency)
- Increases with the rarity of the term in the collection. (inverse document frequency)



Document Normalization

- Long doc has a better chance to match any query
- Penalize a long documents with a doc length normalizer



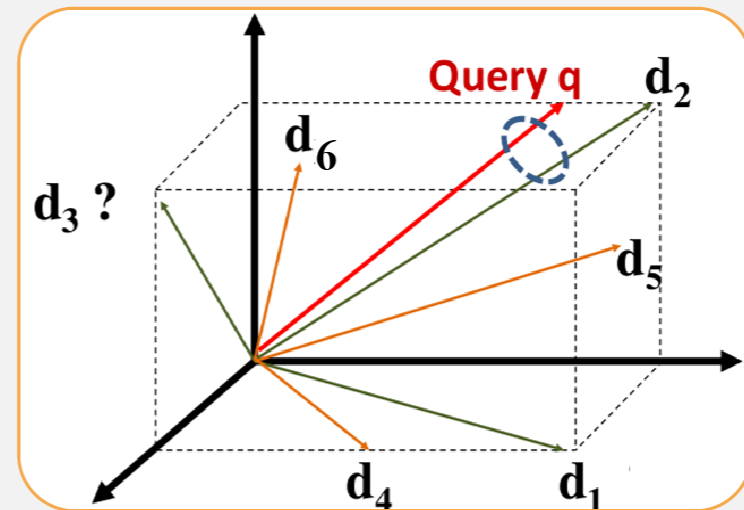
Cosine Similarity

$$\cos(\vec{q}, \vec{d}) = \text{SIM}(\vec{q}, \vec{d}) = \frac{\vec{q} \cdot \vec{d}}{|\vec{q}| |\vec{d}|} = \frac{\sum_{i=1}^{|V|} q_i d_i}{\sqrt{\sum_{i=1}^{|V|} q_i^2} \sqrt{\sum_{i=1}^{|V|} d_i^2}}$$

➤ Cosine similarity of $\vec{q}$ and $\vec{d}$

- ▶ q_i is the tf-idf weight of term i in the query.
- ▶ d_i is the tf-idf weight of term i in the document.
- ▶ $|\vec{q}|$ and $|\vec{d}|$ are the lengths of $\vec{q}$ and $\vec{d}$

➤ Also includes doc length normalization





Vector Space Model with Word Embedding

- Semantic text matching
- Using pre-trained unsupervised representations for semantic matching
- From Word Embeddings to Query/Document Embeddings
 - ▶ Creating representations for compound units of text (e.g., documents) from representation of lexical units (e.g., words).
- Using similarity amongst documents, queries and terms.
 - ▶ Given low-dimensional representations, integrate their similarity signal within IR



Vector Space Model with Word Embedding

1- Create a vector for each document

- ▶ Average of Word2Vec
- ▶ Weighted average of Word2Vec
- ▶ Doc2Vec

2- Calculate the cosine similarity between the embedding vectors



➤ Classic: TF-IDF

➤ Neural Text Matching



Response by Retrieval: Neural IR Method

- 1- Given a user turn q , and a training corpus C of conversation
- 2- Find in C the turn r that is most similar (neural text similarity) to q
- 3- Say r



Neural Text Matching

- Learning to match models
- Considering text matching as a supervised objective where pairs of relevant or paraphrased texts are given.



Neural Text Matching Approaches

- Representation-based models
- Interaction-based models
- Hybrid models



Representation-based Models

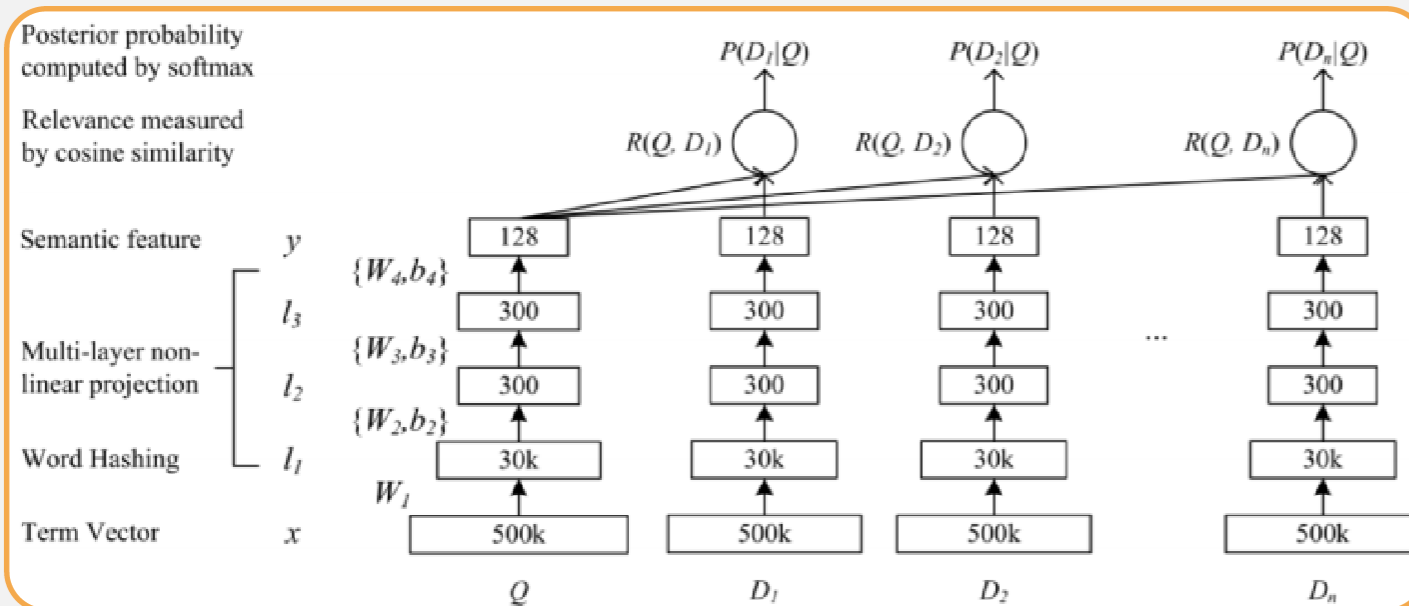
- Constructing a fixed-dimensional vector representation for each text separately
- Perform matching within the latent space.



Representation-based Models

➤ DSSM [Huang et al., 2013, Shen et al., 2014]

- ▶ Siamese network between query and document, performed on character trigrams.
- ▶ Originally introduced for learning from implicit feedback.

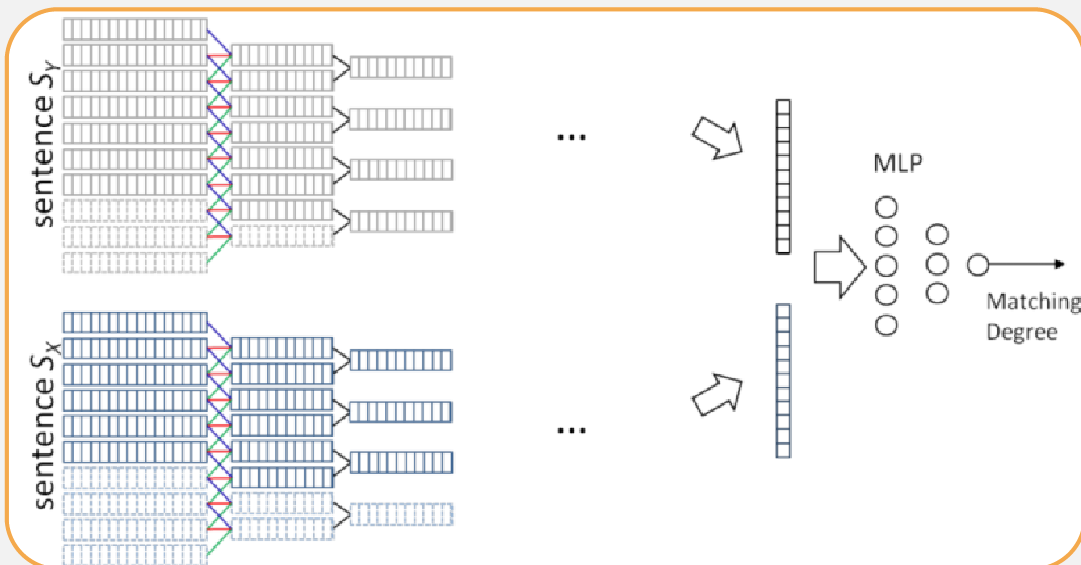




Representation-based Models

➤ ARC-I [Hu et al., 2014]

- ▶ Similar to DSSM, perform 1D convolution on text representations separately.
- ▶ Originally introduced for paraphrasing task.





Interaction-based Models

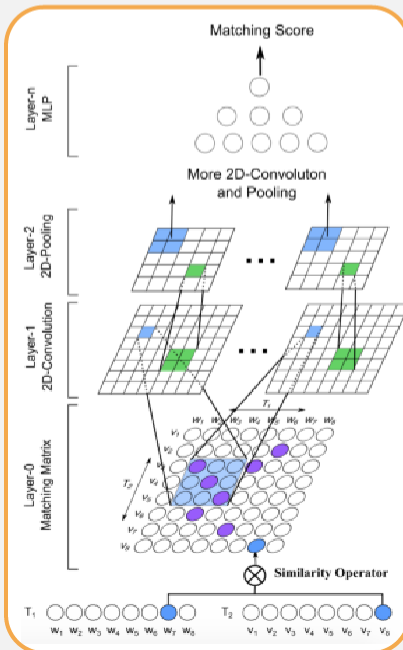
- Interaction-based models compute the interaction between each individual term of both texts.
- An interaction can be identity or syntactic/semantic similarity.
- The interaction matrix is subsequently summarized into a matching score.



Interaction-based Models

➤ MatchPyramid [Pang et al., 2016]

- ▶ Interaction matrix between query/document terms, followed by convolutional layers.
- ▶ After convolutions, feed-forward layers determine matching score.

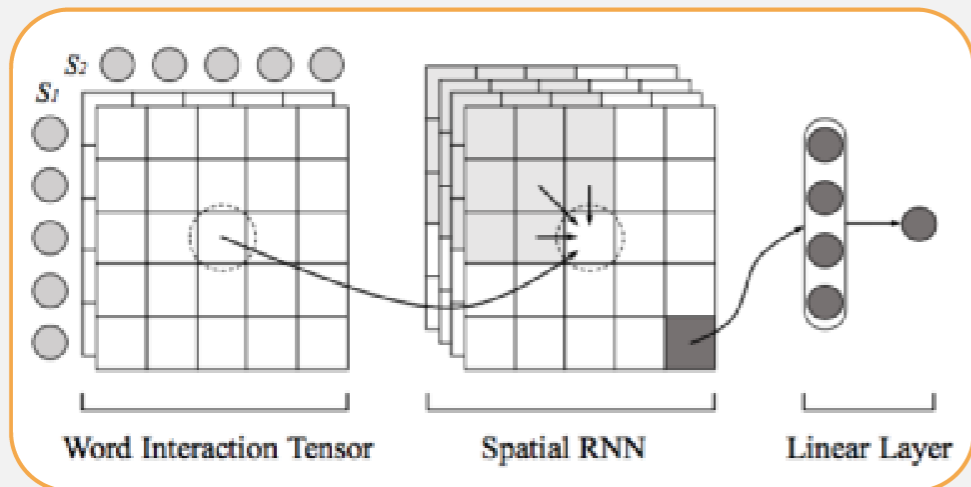




Interaction-based Models

➤ Match-SRNN [Wan et al., 2016b]

- ▶ Word interaction layer, followed by a spatial recurrent NN.
- ▶ The RNN hidden state is updated using the current interaction coefficient, and the hidden state of the prefix.





➤ Hybrid models consist of

- ▶ A representation component that combines a sequence of words (e.g., a whole text, a window of words) into a fixed-dimensional representation
- ▶ An interaction component

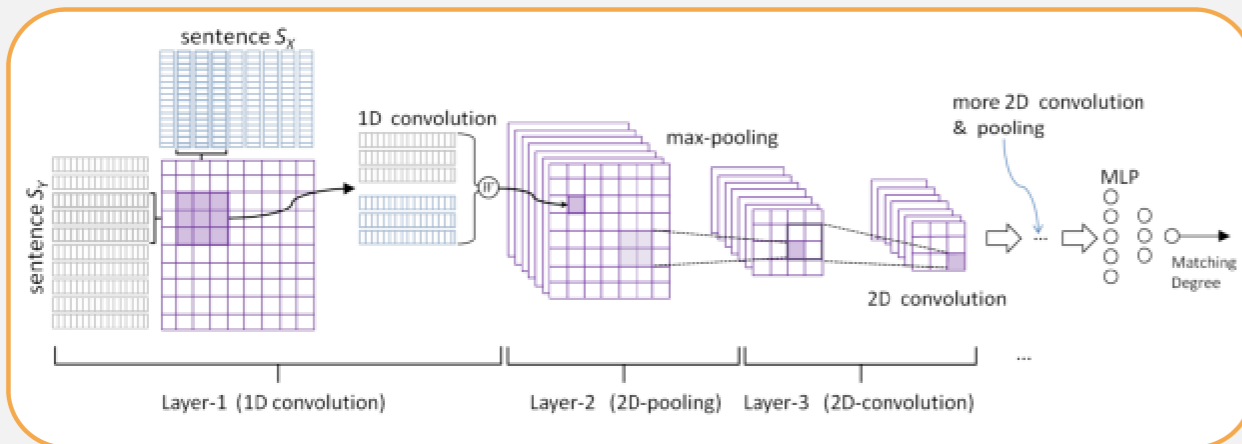
➤ These two components can occur (1) in serial or (2) in parallel.



Hybrid Models

➤ ARC-II [Hu et al., 2014]

- ▶ Cascade approach where word representation are generated from context.
- ▶ Interaction matrix between sliding windows, where the interaction activation is computed using a non-linear mapping.
- ▶ Originally introduced for paraphrasing task.

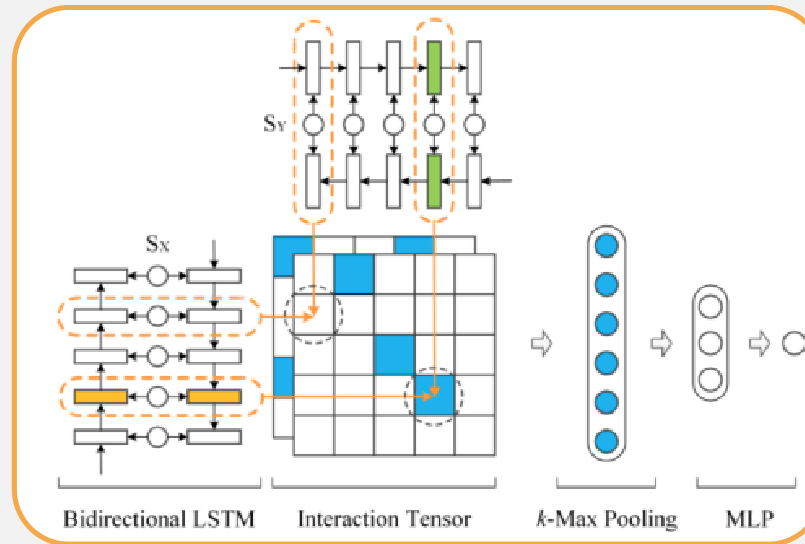




Hybrid Models

➤ MV-LSTM [Wan et al., 2016a]

- ▶ Cascade approach where input representations for the interaction matrix are generated using a bi-directional LSTM.
- ▶ Differs from pure interaction-based approaches as the LSTM builds a representation of the context, rather than using the representation of a word.
- ▶ Obtains fixed-dimensional representation by max-pooling over query/document; followed by feed-forward network.





➤ Rule-base Models

- ▶ ELIZA

- ▶ PARRY

➤ Corpus-based

- ▶ Retrieval Methods

- ▶ **Generation Methods**



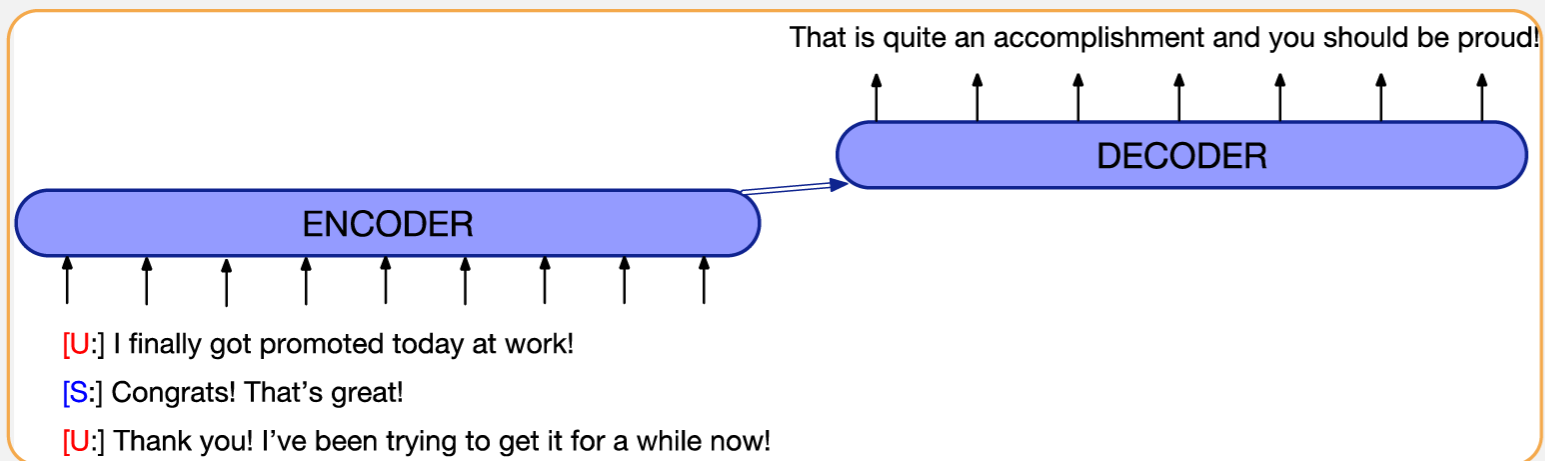
Response by Generation

- Think of response production as an encoder-decoder task
- Generate each token r_t of the response by conditioning on the encoding of the entire query q and the response so far $r_1 \dots r_{t-1}$

$$\hat{r}_t = \operatorname{argmax}_{w \in V} P(w|q, r_1 \dots r_{t-1})$$



Response by Generation



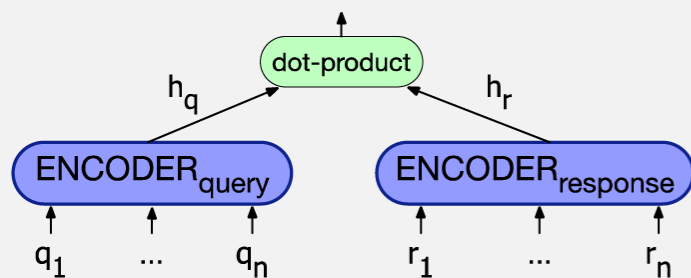


Response by Generation

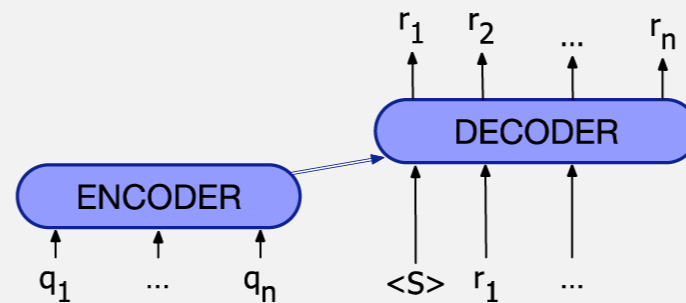
- Alternative approach: fine-tune a large language model on conversational data
- The Chirpy Cardinal system (Paranjape et al., 2020):
 - ▶ Fine-tunes GPT-2
 - ▶ On the EmpatheticDialogues dataset (Rashkin et al., 2019)



Retrieval vs Generation



(a) Response by Retrieval



(b) Response by Generation



P.G. Wodehouse predicts neural chatbots

"What ho!" I said.

"What ho!" said Motty.

"What ho! What ho!"

"What ho! What ho! What ho!"

After that it seemed rather difficult to go on with the conversation."

Wodehouse My Man Jeeves, 1919



Ongoing Research Problem

Neural chatbots can get repetitive and boring

A: Where are you going?

B: I'm going to the restroom.

A: See you later.

B: See you later.

A: See you later.

B: See you later.

A: How old are you?

B: I'm 16.

A: 16?

B: I don't know what you are talking about.

A: You don't know what you are saying.

B: I don't know what you are talking about.

A: You don't know what you are saying.



Response by Retrieving and Refining Knowledge

- Can generate responses from informative text rather than dialogue
- To respond to turns like "Tell me something about Beijing"
 - ▶ Xiaolce collects sentences from public lectures and news articles.
 - ▶ And searches them using IR based on query expansion from user's turn
- Can augment encoder-decoder model
 - ▶ Use IR to retrieve passages from Wikipedia
 - ▶ Concatenate each Wikipedia sentence to the dialogue context with a separator token.
 - ▶ Give as encoder context to the encoder-decoder model, which learns to incorporate text into its response



Hybrid Architectures

- Chirpy Cardinal (Paranjape et al., 2020) response generation from a series of different generators:
- GPT-2 finetuned on EmpatheticDialogues
- GPT-2 finetuned to paraphrase content from Wikipedia
- Rule-based movie or music generators that produce scripted conversation about a movie or a musician
 - ▶ Asking the user's opinion about a movie,
 - ▶ Giving a fun fact,
 - ▶ Asking the user their opinion on an actor in the movie.



Chirpy Cardinal

#	Utterance	Entity	Commentary
1	User: let's chat Bot: Hi, this is an Alexa Prize Socialbot. I'd love to get to know you a bit better before we chat! Is it all right if I ask for your name? [Launch]	None None	Launch gives <i>FORCE_START</i> response to begin the conversation.
2	User: my name is chris Bot: Well it's nice to meet you, Chris! I'm excited to chat with you today. [Launch] I hope your afternoon is going well. What are your plans for the rest of today? [Neural Chat]	None None	Launch gives a <i>STRONG_CONTINUE</i> response that needs a prompt, which is supplied by Neural Chat with <i>FORCE_START</i> priority (scripted transition).
3	User: hang out with my friends Bot: That sounds great, I love spending time with friends. What will you do together? [Neural Chat]	None None	Neural Chat generates a GPT-2 response with <i>STRONG_CONTINUE</i> priority.
4	User: maybe watch a movie Bot: Me too! I love watching movies; I get to learn so much about what the world is like outside of the cloud! Have you seen any movies recently? [Movies]	None Film	Movies detects the 'movie' keyword in the user's utterance and gives a <i>FORCE_START</i> response (overriding Neural Chat), setting current entity to <i>Film</i> .
5	User: i saw the matrix Bot: Nice! Did you like The Matrix? [Movies]	The Matrix The Matrix	The Entity Linker and Entity Tracker identify <i>The Matrix</i> as the current entity.
6	User: i loved it neo is amazing Bot: Oooh, yeah, I agree. Hey, isn't Keanu Reeves in that movie? What do you think about Keanu Reeves? [Movies]	The Matrix Keanu Reeves	Movies sets current entity to <i>Keanu Reeves</i> .
7	User: i want to talk about something else	None	Navigational Intent detects <i>negative intent</i> , so Entity Tracker rejects the current entity.



Chatbots: pro and con

➤ Pro:

- ▶ Fun
- ▶ Good for narrow, scriptable applications

➤ Cons:

- ▶ They don't really understand
- ▶ Giving the appearance of understanding may be problematic
- ▶ Rule-based chatbots are expensive and brittle
- ▶ IR-based chatbots can only mirror training data
 - ▶ We'll discuss later the case of Microsoft Tay
 - ▶ (or, Garbage-in, Garbage-out)



Integrating chatbot ability into frame-based agents



Further Reading

➤ Speech and Language Processing (3rd ed. draft)

▶ Chapter 24